

```

Processando config 11, seed 1, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

```

```

smultin_MultinomialBitsOver test:

```

```

-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

```

```

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

```

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

```

```

-----
Test Results for Delta = 1.0000

```

```

Number of degrees of freedom : 2
Value of the statistic : 2.35
p-value of test : 0.31

```

```

-----
CPU time used : 00:00:00.03

```

Generator state:

16777216 bits have been read.

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

```

```

smultin_MultinomialBitsOver test:

```

```

-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

```

```

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 8.48  
p-value of test : 0.39

-----  
CPU time used : 00:00:00.03

Generator state:

33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 138.19  
p-value of test : 0.25

-----  
CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32765.83  
p-value of test : 0.50

-----  
CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 188.61  
p-value of test : 0.26

-----  
CPU time used : 00:00:00.03

Generator state:

83886080 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 291.37  
p-value of test : 0.70  
-----

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.66  
p-value of test : 0.26  
-----

CPU time used : 00:00:00.01

Generator state:

117440512 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 33.87 |
| p-value of test              | : | 0.38  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 41.47 |
| p-value of test              | : | 0.10  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 19.95 |
| p-value of test              | : | 0.95  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 29.59 |
| p-value of test              | : | 0.24  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 15.97 |
| p-value of test              | : | 0.38  |

-----  
CPU time used : 00:00:00.07

Generator state:

134217728 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 72.34 |
| p-value of test              | : | 0.17  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 56.14 |
| p-value of test              | : | 0.69  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 153.38 |
| p-value of test              | : | 0.63   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 44.67 |
| p-value of test              | : | 0.88  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 39.57 |
| p-value of test              | : | 0.17  |

-----  
CPU time used : 00:00:00.06

Generator state:  
150994944 bits have been read.

===== Summary results of Alhabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.32                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 0.70
p-value of test                   : 0.70
```

```
-----
CPU time used                     : 00:00:00.01
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 8    |
| Value of the statistic       | : | 3.97 |
| p-value of test              | : | 0.86 |

-----  
CPU time used : 00:00:00.01

Generator state:

184549376 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 126.71 |
| p-value of test              | : | 0.52   |

-----  
CPU time used : 00:00:00.01

Generator state:

201326592 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 0.00097653271, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 32768
Value of the statistic : 32979.38
p-value of test : 0.20
```

```
-----
CPU time used : 00:00:00.03
```

Generator state:

218103808 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

sstring\_HammingIndep test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom : 177
Chi-square statistic : 240.56
p-value of test : 1.0e-3
```

```
-----
CPU time used : 00:00:00.01
```

Generator state:

234881024 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 289.55 |
| p-value of test              | : | 0.73   |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.03 |
|---------------|---|-------------|

Generator state:

251658240 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |   |      |
|------------------|---|------|
| Normal statistic | : | 0.57 |
| p-value of test  | : | 0.28 |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.01 |
|---------------|---|-------------|

Generator state:

268435456 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 36.70 |
| p-value of test              | : | 0.26  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 30.95 |
| p-value of test              | : | 0.47  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 28.79 |
| p-value of test              | : | 0.63  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 32.07 |
| p-value of test              | : | 0.16  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 19.48 |
| p-value of test              | : | 0.19  |

-----  
CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 52.55 |
| p-value of test              | : | 0.80  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 71.58 |
| p-value of test              | : | 0.19  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 199.65 |
| p-value of test              | : | 0.02   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 40.47 |
| p-value of test              | : | 0.95  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 26.48 |
| p-value of test              | : | 0.74  |

-----  
CPU time used : 00:00:00.06

Generator state:

301989888 bits have been read.

===== Summary results of Alphabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufire_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.28                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 1.47
p-value of test                   : 0.48
```

```
-----
CPU time used                      : 00:00:00.01
```

Generator state:

318767104 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
```

Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

$\Delta = 1$ ,  $\mu = 2.0861626e-07$ ,  $\sigma = 1$

-----  
Test Results for  $\Delta = 1.0000$

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 8    |
| Value of the statistic       | : | 7.61 |
| p-value of test              | : | 0.47 |

-----  
CPU time used : 00:00:00.01

Generator state:  
335544320 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

$\Delta = 1$ ,  $\mu = 3.784895e-06$ ,  $\sigma = 1$

-----  
Test Results for  $\Delta = 1.0000$

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 108.48 |
| p-value of test              | : | 0.89   |

-----  
CPU time used : 00:00:00.01

Generator state:  
352321536 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32734.27  
p-value of test : 0.55

-----  
CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 174.33  
p-value of test : 0.54

-----  
CPU time used : 00:00:00.03

Generator state:

385875968 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 268.04  
p-value of test : 0.94  
-----

CPU time used : 00:00:00.01

Generator state:

402653184 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 1.22  
p-value of test : 0.11  
-----

CPU time used : 00:00:00.01

Generator state:

419430400 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 30.24 |
| p-value of test              | : | 0.56  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 32.78 |
| p-value of test              | : | 0.38  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 29.76 |
| p-value of test              | : | 0.58  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 31.08 |
| p-value of test              | : | 0.19  |

-----  
Test on the values of the Statistic C

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 15   |
| ChiSquare statistic          | : | 5.74 |
| p-value of test              | : | 0.98 |

-----  
CPU time used : 00:00:00.06

Generator state:

436207616 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 63.39 |
| p-value of test              | : | 0.43  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 63.25 |
| p-value of test              | : | 0.43  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 161.30 |
| p-value of test              | : | 0.46   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 51.81 |
| p-value of test              | : | 0.67  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 29.83 |
| p-value of test              | : | 0.58  |

-----  
CPU time used : 00:00:00.04

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

|                 |                                                |
|-----------------|------------------------------------------------|
| Version:        | TestU01 1.2.3                                  |
| Generator:      | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits: | 16777216                                       |

Number of statistics: 17  
Total CPU time: 00:00:00.26

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alhabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 2    |
| Value of the statistic       | : | 0.90 |
| p-value of test              | : | 0.64 |

-----  
CPU time used : 00:00:00.01

Generator state:  
469761792 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 3.94
p-value of test              : 0.86
```

-----  
CPU time used : 00:00:00.01

Generator state:

486539008 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 120.43
p-value of test              : 0.67
```

-----  
CPU time used : 00:00:00.01

Generator state:

503316224 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32824.68  
p-value of test : 0.41

-----

CPU time used : 00:00:00.03

Generator state:

520093440 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 184.82  
p-value of test : 0.33

-----

CPU time used : 00:00:00.01

Generator state:  
536870656 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 296.99 |
| p-value of test              | : | 0.62   |

-----  
CPU time used : 00:00:00.01

Generator state:  
553647872 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.73  
p-value of test : 0.23

-----  
CPU time used : 00:00:00.01

Generator state:  
570425088 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 25.87 |
| p-value of test              | : | 0.77  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 27.20 |
| p-value of test              | : | 0.66  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 30.13 |
| p-value of test              | : | 0.56  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 27.34 |
| p-value of test              | : | 0.34  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 10.38 |
| p-value of test              | : | 0.80  |

-----  
CPU time used : 00:00:00.06

Generator state:

587202304 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config11_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 70.48 |
| p-value of test              | : | 0.22  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 66.34 |
| p-value of test              | : | 0.33  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 155.32 |
| p-value of test              | : | 0.59   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 50.99 |
| p-value of test              | : | 0.70  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 35.13 |
| p-value of test              | : | 0.32  |

-----  
CPU time used : 00:00:00.04

Generator state:  
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 3.32  
p-value of test : 0.19

-----  
CPU time used : 00:00:00.01

Generator state:  
620756480 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 4.58  
p-value of test : 0.80

-----  
CPU time used : 00:00:00.01

Generator state:  
637533696 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 127.01  
p-value of test : 0.51

-----  
CPU time used : 00:00:00.01

Generator state:  
654310912 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32605.59  
p-value of test : 0.74

-----

CPU time used : 00:00:00.04

Generator state:

671088128 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 162.34  
p-value of test : 0.78

-----

CPU time used : 00:00:00.01

Generator state:

687865344 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 305

Chi-square statistic : 316.91

p-value of test : 0.31

-----

CPU time used : 00:00:00.01

Generator state:

704642560 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

Normal statistic : 1.75

p-value of test : 0.04

-----

CPU time used : 00:00:00.01

Generator state:

721419776 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 35.90 |
| p-value of test              | : | 0.29  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 32.65 |
| p-value of test              | : | 0.39  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 47.77 |
| p-value of test              | : | 0.04  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 12.17 |
| p-value of test              | : | 0.99  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 14.85 |
| p-value of test              | : | 0.46  |

-----  
CPU time used : 00:00:00.06

Generator state:  
738196992 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 69.53 |
| p-value of test              | : | 0.24  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 82.41 |
| p-value of test              | : | 0.04  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 142.17 |
| p-value of test              | : | 0.84   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 45.40 |
| p-value of test              | : | 0.87  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 19.57 |
| p-value of test              | : | 0.96  |

-----  
CPU time used : 00:00:00.04

Generator state:  
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 1.01  
p-value of test : 0.60

-----

CPU time used : 00:00:00.03

Generator state:

771751168 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 7.42  
p-value of test : 0.49

-----  
CPU time used : 00:00:00.01

Generator state:

788528384 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 131.37  
p-value of test : 0.40

-----  
CPU time used : 00:00:00.01

Generator state:  
805305600 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32628.23  
p-value of test : 0.71

-----

CPU time used : 00:00:00.01

Generator state:  
822082816 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 191.37  
p-value of test : 0.22

-----  
CPU time used : 00:00:00.01

Generator state:  
838860032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 303.68  
p-value of test : 0.51

-----  
CPU time used : 00:00:00.01

Generator state:  
855637248 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -0.48  
p-value of test : 0.68

-----  
CPU time used : 00:00:00.03

Generator state:  
872414464 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 25.76 |
| p-value of test              | : | 0.77  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 29.48 |
| p-value of test              | : | 0.54  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 24.11 |
| p-value of test              | : | 0.84  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 35.32 |
| p-value of test              | : | 0.08  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 26.93 |
| p-value of test              | : | 0.03  |

-----  
CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config11\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 60.22 |
| p-value of test              | : | 0.54  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 51.08 |
| p-value of test              | : | 0.84  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 164.61 |
| p-value of test              | : | 0.38   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 36.15 |
| p-value of test              | : | 0.99  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 30.40 |
| p-value of test              | : | 0.55  |

-----  
CPU time used : 00:00:00.06

Generator state:

905968640 bits have been read.

===== Summary results of Alphabit =====

Version:            TestU01 1.2.3  
Generator:          ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits:     16777216  
Number of statistics: 17  
Total CPU time:     00:00:00.26

All tests were passed